



Multi-Agent Systems Quiz



Section A: MCQs (Conceptual Understanding)

1. What is the primary difference between a single-agent system and a multi-agent system?

- A. Single-agent systems use AI, multi-agent systems do not
- B. Multi-agent systems involve multiple interacting agents
- C. Single-agent systems are always faster
- D. Multi-agent systems cannot make decisions independently

Answer: B. Multi-agent systems involve multiple interacting agents

2. In a multi-agent system, agents are typically:

- A. Completely dependent on each other
- B. Independent but capable of interaction
- C. Controlled by a central authority only
- D. Static and non-adaptive

Answer: B. Independent but capable of interaction

3. Which of the following is an example of a multi-agent system?

- A. Calculator app
- B. Self-driving car network communicating with other cars
- C. Text editor
- D. Single chatbot

Answer: B. Self-driving car network communicating with other cars

4. What is a key advantage of multi-agent systems?

- A. Simpler design
- B. No communication required
- C. Scalability and parallelism
- D. Limited decision-making

Answer: C. Scalability and parallelism

5. Which architecture uses only one decision-making entity?

- A. Distributed system
- B. Multi-agent system
- C. Single-agent system
- D. Hybrid system

Answer: C. Single-agent system

Section B: Short Answer Questions

6. Define a multi-agent system in your own words.

Answer: A multi-agent system (MAS) is a system where multiple intelligent agents interact, cooperate, or compete to achieve goals.

7. List two advantages of multi-agent systems over single-agent systems.

Answer: Two advantages:

- **Scalability (can grow easily)**
- **Parallel problem solving (faster execution)**

8. What is agent communication, and why is it important?

Answer: Agent communication is the exchange of information between agents. Important because it enables coordination, cooperation, and decision-making.

9. Name two types of agent communication patterns.

Answer: Two communication patterns:

- **Peer-to-peer**
- **Broadcasting**

10. What is task delegation in multi-agent systems?

Answer: Task delegation is when one agent assigns tasks to others based on their abilities.

Section C: Analytical / Application-Based Questions

11. Compare single-agent and multi-agent systems based on:

- Decision-making
- Scalability
- Complexity

Answer:

• Aspect	• Single-Agent	• Multi-Agent
• Decision-making	• Centralized	• Distributed
• Scalability	• Limited	• High
• Complexity	• Low	• High
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12. Explain the difference between **centralized** and **decentralized** multi-agent architectures.

Answer: Centralized vs Decentralized:

- **Centralized: One controller manages all agents**
Easier control
Single point of failure
 - **Decentralized: Agents act independently**
Robust & scalable
More coordination complexity
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13. A group of delivery robots must coordinate to deliver packages efficiently.

- What type of system is this?
- Why is communication important here?

Answer: ☐ **Type: Multi-agent system**

- ☐ **Communication importance:**
- **Avoid collisions**
- **Optimize routes**

- **Share workload**
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14. If one agent assigns tasks to others based on their capabilities, what strategy is being used?

- A. Broadcasting
- B. Task delegation
- C. Isolation
- D. Random assignment

Answer: B. Task delegation

15. Explain how **peer-to-peer communication** works in multi-agent systems.

Answer: Agents communicate directly with each other without a central controller.

Improves flexibility and reduces dependency.



Section D: Scenario-Based Questions

16. In a smart traffic system, traffic lights (agents) communicate with each other to reduce congestion.

- Identify the communication pattern used
- What could happen if communication fails?

Answer: Smart traffic system:

- **Pattern: Peer-to-peer communication**
 - **If communication fails:**
 - **Traffic congestion increases**
 - **Poor coordination**
 - **Possible accidents**
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17. Consider a team of AI agents solving a complex problem:

- One agent plans
- Others execute tasks

What architecture or strategy is this? Explain.

Answer: Hierarchical architecture / task delegation strategy

- **One agent = planner**
 - **Others = executors**
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18. A multi-agent system uses broadcasting to share information.

- What are the advantages and disadvantages of this approach?

Answer: Advantages:

- **Fast information sharing**
 - **Simple implementation**
 - **Disadvantages:**
 - **Network overload**
 - **Irrelevant data to some agents**
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19. In a distributed AI system, each agent has partial knowledge of the environment.

- How can agents collaborate effectively?

Answer: Collaboration with partial knowledge:

- **Share information**
 - **Use coordination protocols**
 - **Combine local knowledge → global solution**
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20. Suppose a system fails when one central controller crashes.

- What type of architecture is this?
- Suggest a better alternative.

Answer: ☐ **Type: Centralized architecture**

- ☐ **Better alternative: Decentralized system (no single failure point)**
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Section E: Higher-Order Thinking

21. Design a simple multi-agent system for online food delivery.

- Define roles of at least 3 agents

Answer: Example agents:

- **Order Agent** → Handles customer orders
 - **Delivery Agent** → Assigns riders & routes
 - **Restaurant Agent** → Prepares food
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22. Why is task delegation important in large-scale systems like cloud computing?

Answer: Importance of task delegation:

- **Efficient resource use**
 - **Faster processing**
 - **Handles large workloads (e.g., cloud systems)**
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23. Compare cooperative vs competitive agents with examples.

Answer: Cooperative vs Competitive agents:

- **Cooperative: Work together (e.g., robots assembling a product)**
- **Competitive: Compete for goals/resources (e.g., bidding systems)**

24. How does communication overhead impact system performance?

Answer: Communication overhead impact:

- **Too much communication → slows system**
- **Uses bandwidth & processing power**
Must be optimized

25. Suggest improvements for a poorly coordinated multi-agent system.

Answer: Improvements for poor coordination:

- **Better communication protocols**
- **Clear task allocation**
- **Use decentralized control**
- **Improve agent learning/adaptation**